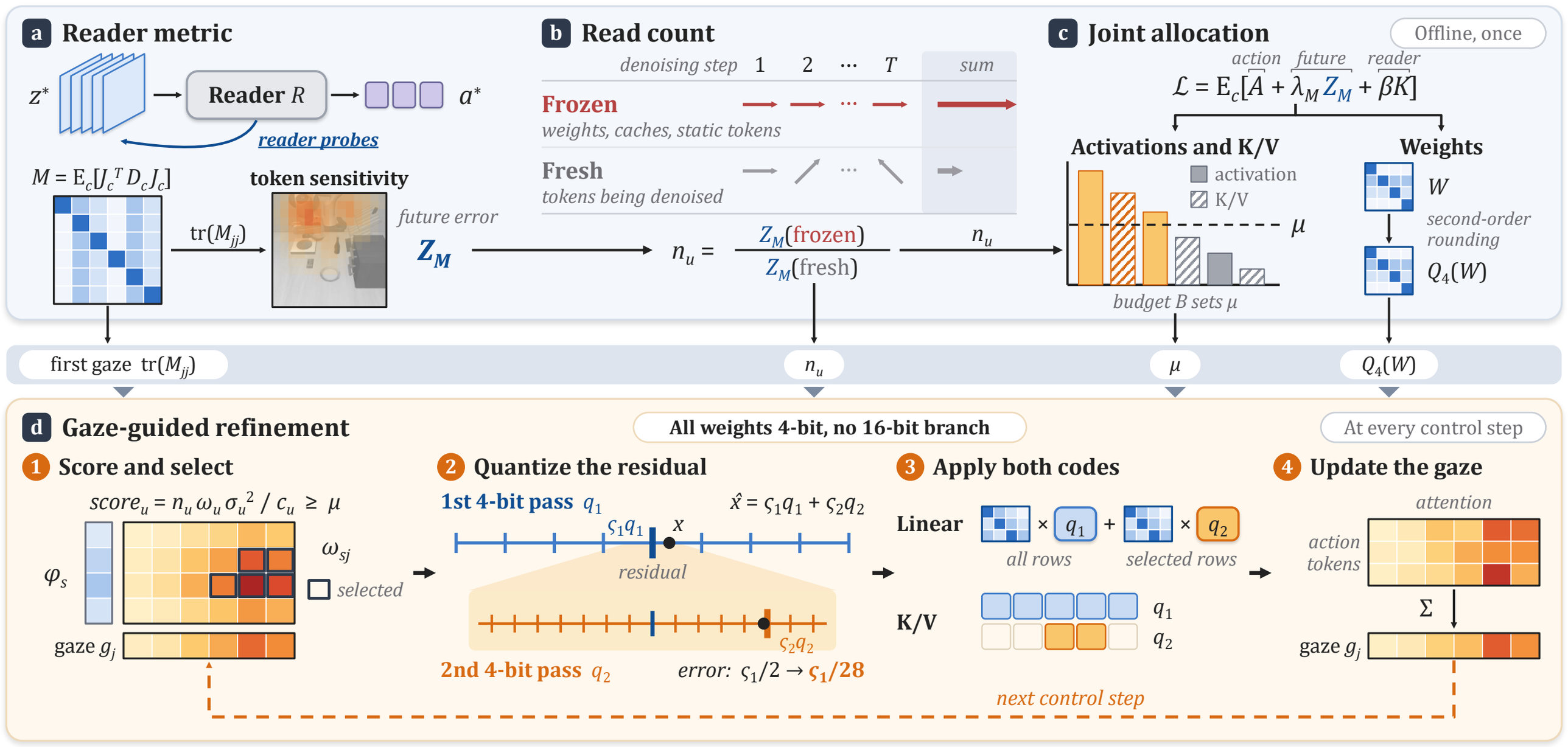

a Reader metric
z^*
Reader R
a^*
reader probes
$M = \mathrm{E}_c[J_c^T D_c J_c]$
token sensitivity
$\mathrm{tr}(M_{jj})$
future error
Z_M
b Read count
denoising step 1 2 ⋯ T
sum
Frozen
weights, caches, static tokens
Fresh
tokens being denoised
$n_u = \frac{Z_M(\text{frozen})}{Z_M(\text{fresh})}$
n_u
c Joint allocation
Offline, once
action future reader
$\mathcal{L} = \mathrm{E}_c[A + \lambda_M Z_M + \beta K]$
Activations and K/V
Weights
activation
K/V
μ
budget B sets μ
W
second-order rounding
$Q_4(W)$
first gaze $\mathrm{tr}(M_{jj})$
n_u
μ
$Q_4(W)$
d Gaze-guided refinement
All weights 4-bit, no 16-bit branch
At every control step
1 Score and select
$score_u = n_u \omega_u \sigma_u^2 / c_u \geq \mu$
φ_s
ω_{sj}
selected
gaze g_j
2 Quantize the residual
1st 4-bit pass q_1
$\hat{x} = \varsigma_1 q_1 + \varsigma_2 q_2$
$\varsigma_1 q_1$
x
residual
$\varsigma_2 q_2$
2nd 4-bit pass q_2
error: $\varsigma_1/2 \rightarrow \varsigma_1/28$
3 Apply both codes
Linear
all rows
selected rows
K/V
q_1
q_2
4 Update the gaze
attention
action tokens
Σ
gaze g_j
next control step